Non-Context-Free language

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- There arises a question, whether all languages are CFL? The answer is no.
- Languages which are not Context-Free are called Non-CFL.
- To prove the claim that all languages are not Context-Free, the study of machines of word production from the grammar is needed

Terminology

- **Live production**:

  Nonterminal → Nonterminals Nonterminals

- **Dead production:**

  Nonterminal → terminal

- It may be noted that every CFG in CNF has only these types of productions.

Theorem 32

- **If a CFG is in CNF and if there is restriction to use the live production at most once each, then only the finite many words can be generated.**
- It may be noted that every time a live production is applied during the derivation of a word it increases the number of non-terminals by one.
- Similarly applying dead production decreases the non-terminals by one. Which shows that to generate a word, one more dead production are applied than the live productions *e.g.*

Example

- S => XY
  => aY
  => ab

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- S => XY
  => aY
  => ab
- Here one live and two dead productions are used to generate a word ab.

Proof of Theorem 32

- Suppose we start in some abstract CFG in CNF with

$$S \Rightarrow AB$$

- Suppose we apply the live production $A \rightarrow XY$, we get

$$\Rightarrow XYB$$

which has 3 nonterminals (i.e., one more nonterminal).

- If we now apply the dead production $X \rightarrow b$, we obtain

$$\Rightarrow bYB$$

which has 2 nonterminals (i.e., one less nonterminal).

Proof of Theorem 32

- Thus, every time we apply a live production, we increase the number of nonterminal by one. Every time we apply a dead production, we decrease the number of nonterminals by one.
- Hence, to start from a nonterminal S and eventually arrive at a string of only terminals (i.e., no nonterminals), we must **apply one more dead production than live production**.
- Suppose the the grammar G has exactly p live productions and q dead productions.
- Then, any derivation that does not re-use a live production must have at most p live productions, and accordingly at most $(p + 1)$ dead productions.

Proof of Theorem 32

- Since each letter in the final word comes from one dead production, all words generated from G without repeating any live productions have at most $(p+1)$ letters in them.
- Since no words can be more than $(p+1)$ letters long, there can only be finitely many words.